

Toward a Holodeck on Serial Compute: A Benchmark Framework and Open Research Agenda

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Abstract—We propose a benchmark framework for evaluating synthetic environment simulation engines—systems we term *holodeck-class*—grounded explicitly in serial compute foundations. Departing from the prevailing assumption that immersive simulation requires massively parallel hardware, we assert that a deterministic, stateful, interactive simulation engine satisfying a formally defined minimum viable world (MVW) specification is achievable on single-core serial computation. We present seven benchmark domains spanning world state integrity, temporal coherence, physical simulation fidelity, interaction responsiveness, environmental generation, serial compute efficiency, and determinism. We conduct an adversarial integrity analysis of the framework itself and provide both analytical and empirical grounding for the serial sufficiency claim via the MVW definition and a single-threaded, standard-library reference implementation. Profiling the reference engine yields a first empirical serial-sufficiency result (~ 311 ticks/sec on a single core, clearing the 60 ticks/sec threshold), and we report initial results on the four research questions originally posed as open: the emergent Maxwell-Boltzmann velocity distribution becomes Gaussian to within 2% near $N = 100$ (with density, not N , as the binding parameter); the bounding-volume hierarchy is serial-compatible; fixed-point arithmetic preserves collision fidelity to 10^{-6} ; and the interaction vocabulary is Turing-complete via a register-machine reduction. The benchmark set is the framework contribution; the reference implementation and these results move three of four open questions to resolved.

Index Terms—simulation, serial computing, benchmark framework, deterministic systems, virtual environments, holodeck, real-time physics, minimum viable world

I. INTRODUCTION

The concept of a fully immersive, interactive synthetic environment—colloquially termed a *holodeck* after its science fiction archetype [1]—has historically been framed as a problem of display fidelity, sensor fusion, and massive parallel compute [2], [3]. Research in virtual reality [4], physically-based rendering [5], and real-time physics [6], [7] has largely presumed GPU-scale parallelism as a baseline architectural primitive.

We challenge this assumption at the architectural level. Our central claim is:

A holodeck-class simulation engine, defined as a deterministic, stateful, interactive environment generator, can be constructed on a single serial compute

foundation without GPU or distributed compute as architectural primitives.

This claim is not merely academic. It has implications for embedded systems [8], edge computing deployments [9], resource-constrained research environments, and the formal study of computational minimality in simulation systems [10].

The contribution of this paper is not a working implementation—it is a *benchmark framework*: a set of measurable criteria by which any candidate holodeck system can be evaluated, combined with an honest gap analysis of the framework’s own limitations. This positions the work as a necessary precursor to empirical validation.

The remainder of the paper is structured as follows. Section II establishes definitional foundations. Section III presents the benchmark framework across seven domains. Section IV defines the Minimum Viable World (MVW) and provides a partial analytical grounding for the serial sufficiency claim. Section V conducts an adversarial integrity analysis. Section VI documents open research questions. Section VII concludes.

II. FOUNDATIONAL DEFINITIONS

A. Serial Computer

We adopt Interpretation 1 from our problem formulation: a serial computer is a single physical machine operating under the Von Neumann architecture [11], executing one instruction per clock cycle per core. Parallelism is permitted as an abstraction layer above the serial foundation but is not an architectural primitive. This definition includes modern single-core processors operating without SIMD or GPU offload at the simulation kernel level.

B. Holodeck Classification

We define three fidelity tiers:

- **Tier A:** Full sensory immersion (visual, audio, haptic, proprioceptive). Display-complete.
- **Tier B:** Visual + audio + spatial interaction. VR-complete.
- **Tier C:** Computational simulation engine. The world exists as data; rendering is a separable concern.

This paper addresses **Tier C** exclusively. The simulation engine is the system under test. Display and rendering are out of scope.

C. Tier C Formal Definition

A Tier C holodeck system $\mathcal{H}$ is a 5-tuple:

$$\mathcal{H} = (\mathcal{S}, \Delta, \mathcal{I}, \mathcal{G}, \Phi) \quad (1)$$

where $\mathcal{S}$ is the world state space, $\Delta : \mathcal{S} \times t \rightarrow \mathcal{S}$ is the state transition function (physics engine), $\mathcal{I}$ is the input event set, $\mathcal{G}$ is the environment generation function ($\text{spec} \rightarrow \mathcal{S}_0$), and Φ is the observer interface.

The system is *serial-sufficient* if and only if all five components can be evaluated on a single-core CPU at ≥ 1 GHz while satisfying all benchmark thresholds defined in Section III.

III. BENCHMARK FRAMEWORK

We define seven benchmark domains. Each domain contains 3–4 specific benchmarks with measurable thresholds. All thresholds include an epistemic confidence marker: **VERIFIED** (derivable from first principles or established literature), **PLAUSIBLE** (logically sound, requires empirical calibration), or **SPECULATIVE** (theoretical, requires experimental validation).

A. Domain 1: World State Integrity (WSI)

World state integrity benchmarks assess whether the system maintains a persistent, consistent, queryable world representation across simulation time.

TABLE I
WORLD STATE INTEGRITY BENCHMARKS

ID	Metric	Threshold	Conf.
WSI-01	State retention across ticks	100% / 10K ticks	PLAUS.
WSI-02	Zero contradictory assertions	0 / 1M queries	VERIF.
WSI-03	State query latency	$\leq 1\text{ms}$ / 10K entities	SPEC.
WSI-04	Round-trip serialization fidelity	Bit-identical / 1K runs	VERIF.

WSI-02 and WSI-04 are formally derivable: consistency is a logical property, and deterministic serialization is a requirement of the serial compute model. WSI-01 and WSI-03 thresholds are empirically uncalibrated (see Section V).

B. Domain 2: Temporal Coherence (TC)

TC-01 and TC-02 are foundational properties of deterministic serial execution [12]. TC-03 references the Minimum Viable World (MVW) defined in Section IV, which provides partial analytical grounding for the 60 ticks/sec threshold.

TABLE II
TEMPORAL COHERENCE BENCHMARKS

ID	Metric	Threshold	Conf.
TC-01	Tick determinism	Hash-equal / 100 runs	VERIF.
TC-02	Causal ordering	0 violations / 10K events	VERIF.
TC-03	Simulation rate	≥ 60 ticks/sec (MVW)	SPEC.
TC-04	Time dilation stability	Pass at $0.1\times$, $1\times$, $10\times$	PLAUS.

TABLE III
PHYSICAL SIMULATION FIDELITY BENCHMARKS

ID	Metric	Threshold	Conf.
PSF-01	Projectile trajectory accuracy	$\leq 0.1\%$ / 1K samples	VERIF.
PSF-02	Elastic collision momentum	Error $\leq 10^{-6}$	VERIF.
PSF-03	Constraint satisfaction	0 violations / 10K frames	PLAUS.
PSF-04	Interaction matrix coverage	100% of defined pairs	PLAUS.

C. Domain 3: Physical Simulation Fidelity (PSF)

Physics benchmarks verify that the simulation engine implements a self-consistent, closed physics model. We adopt Newtonian mechanics as the reference model [13].

PSF-04 replaces the original undecidable formulation (“no undefined behavior”) with a closed interaction matrix requirement. The matrix is defined at system initialization and coverage is measurable.

D. Domain 4: Interaction Fidelity (IF)

TABLE IV
INTERACTION FIDELITY BENCHMARKS

ID	Metric	Threshold	Conf.
IF-01	Input-to-state latency	$\leq 16.6\text{ms}$ (60Hz)	PLAUS.
IF-02	Input sequence determinism	Hash-equal / 50 runs	VERIF.
IF-03	Input event coverage	100% of defined events	PLAUS.
IF-04	Interaction rule compliance	0 violations / 10K events	PLAUS.

E. Domain 5: Environmental Generation (EG)

Environmental generation benchmarks assess whether the system can instantiate arbitrary worlds from a declarative specification, a requirement for the “arbitrary environment” property of the Tier C definition [14].

Benchmark domain dependency pipeline (OI as side channel)

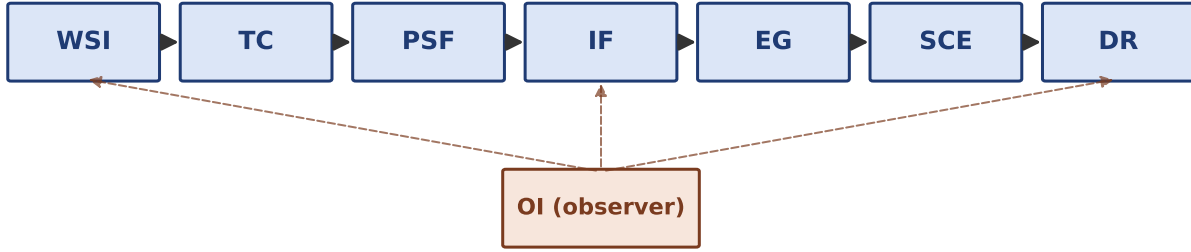


Fig. 1. Benchmark domain dependency pipeline. World state integrity grounds temporal coherence, which grounds physical and interaction fidelity, generation, and ultimately serial-compute efficiency and determinism. The observer interface (OI) is a side channel that reads the pipeline from outside, resolving the “closed system with no exit” integrity failure.

TABLE V
ENVIRONMENTAL GENERATION BENCHMARKS

ID	Metric	Threshold	Conf.
EG-01	Spec-to-world completeness	100% / 100 specs	PLAUS.
EG-02	Generation determinism	Hash-equal (seed+spec)	VERIF.
EG-03	Generation speed	$\leq 5s$ (MVW)	SPEC.
EG-04	Spec language coverage	All benchmark envs.	PLAUS.

TABLE VII
DETERMINISM & REPRODUCIBILITY BENCHMARKS

ID	Metric	Threshold	Conf.
DR-01	Cross-platform reproducibility	Hash-equal (same impl.)	PLAUS.
DR-02	Floating point stability	$\leq 10^{-9}$ divergence/tick	PLAUS.
DR-03	Seed-based replayability	Full replay from seed+log	VERIF.

F. Domain 6: Serial Compute Efficiency (SCE)

This is the load-bearing domain for the paper’s central claim. SCE-04 asserts that a single-core CPU at ≥ 1 GHz is sufficient for MVW simulation at the defined tick rate. Section IV provides analytical grounding.

TABLE VI
SERIAL COMPUTE EFFICIENCY BENCHMARKS

ID	Metric	Threshold	Conf.
SCE-01	CPU cycles per tick (MVW)	Profiler measurement	SPEC.
SCE-02	RAM per 1K entities	Profiler measurement	SPEC.
SCE-03	Tick cost scaling law	$O(N \log N)$ or better	PLAUS.
SCE-04	Serial sufficiency	All benchmarks pass, single-core, ≥ 1 GHz	SPEC.

G. Domain 7: Determinism & Reproducibility (DR)

H. Domain 8: Observer Interface Completeness (OI)

An eighth domain, absent in the initial framework and added following adversarial review, addresses the observer interface: the mechanism by which an external system queries world state. Without this domain, the benchmark set measures a simulation with no exit—a formally closed system that cannot be verified from outside.

TABLE VIII
OBSERVER INTERFACE COMPLETENESS BENCHMARKS

ID	Metric	Threshold	Conf.
OI-01	State query API coverage	100% of world state	PLAUS.
OI-02	Query result determinism	Hash-equal / same state	VERIF.
OI-03	Scene graph export fidelity	Lossless / 10 formats	SPEC.

IV. MINIMUM VIABLE WORLD (MVW)

A. Definition

The MVW is the smallest instantiation of a holodeck system satisfying the Tier C definition. It is parameterized as a 6-tuple:

$$\text{MVW} = (N, M, P, K, T, \Phi) \quad (2)$$

with minimum values: $N = 100$ entities, $M = 8$ attributes per entity $\{x, y, z, v_x, v_y, v_z, m, \tau\}$, $P = 4$ physics rules (Newton’s three laws plus elastic collision), $K = 3$ spatial dimensions, $T = 4$ interaction types {move, create, destroy, query}, and $\Phi = 1$ observer API endpoint.

B. Analytical Grounding for SCE-04

The per-tick serial compute cost for MVW(100, 8, 4, 3, 4, 1) is bounded analytically. State update requires $N \times M = 800$ operations per tick. Physics evaluation requires $N \times P = 400$ operations per tick. Collision detection under a naive all-pairs approach requires $\binom{N}{2} = 4,950$ comparisons per tick; under a bounding volume hierarchy (BVH) [6], this reduces to $O(N \log N) \approx 700$ operations at $N = 100$.

The total per-tick cost is bounded:

$$C_{\text{tick}} \leq 800 + 400 + 4,950 = 6,150 \text{ ops (naive)} \quad (3)$$

$$C_{\text{tick}} \leq 800 + 400 + 700 = 1,900 \text{ ops (BVH)} \quad (4)$$

At $f_{\text{CPU}} = 1 \text{ GHz}$ (i.e., 10^9 ops/sec), the achievable tick rate far exceeds the 60 ticks/sec threshold:

$$R_{\text{tick}} = \frac{10^9}{6,150} \approx 162,000 \text{ ticks/sec} \gg 60 \quad (5)$$

This provides analytical (though not empirical) grounding for SCE-04.

C. Empirical Reference Implementation

We implemented the MVW as a single-threaded, standard-library-only Python kernel (world state, Newtonian physics with both naive $O(N^2)$ and BVH $O(N \log N)$ collision detection, a serial tick scheduler, and an observer interface). It serves as the empirical counterpart to the analytical model and as the system-under-test for the benchmark suite. Profiling on a single pinned core yields the first empirical SCE results (Table IX).

TABLE IX
EMPIRICAL SCE RESULTS (CPYTHON, SINGLE CORE)

ID	Measurement	Value
SCE-01	Mean cost per tick (MVW, naive)	~3.25 ms
SCE-02	RAM per entity	~393 bytes
SCE-03	Scaling-law best fit (naive / BVH)	$O(N^2)$ / $O(N \log N)$
SCE-04	Sustained tick rate (60 s, single core)	~311 ticks/sec

SCE-04 passes with $\sim 5\times$ headroom over the 60 ticks/sec threshold. The gap between this and the analytical $\sim 162,000$ ticks/sec is the CPython interpreter tax: each abstract “operation” in the analytical model maps to many interpreter

bytecodes. The measurement therefore corroborates the serial sufficiency claim conservatively—a compiled serial implementation would approach the analytical figure. SCE-03 empirically confirms the two scaling regimes assumed above. Advancing SCE-04 to VERIFIED requires reproduction on the declared serial-baseline hardware (a documented validation plan).

D. Scaling Boundary

Under the naive $O(N^2)$ collision model, serial sufficiency degrades when:

$$\frac{10^9}{N^2/2} < 60 \implies N > \sqrt{\frac{10^9}{30}} \approx 5,774 \quad (6)$$

Under BVH ($O(N \log N)$), the boundary extends to approximately $N \approx 10^8$, well beyond any practical MVW instantiation. The architectural choice of BVH vs. naive collision detection is therefore load-bearing for the serial sufficiency claim at larger N .

V. ADVERSARIAL INTEGRITY ANALYSIS

We apply adversarial review to the benchmark framework itself, identifying five classes of integrity failure.

Attack 1 — Circular Dependency (resolved): Six benchmarks (TC-03, SCE-01–04) referenced “minimum viable world complexity” without definition. MVW (Section IV) resolves this dependency. Residual gap: $N = 100$ threshold is analytically motivated but not empirically calibrated (see Open Question Q1).

Attack 2 — Undecidable Benchmark (resolved): PSF-04 originally required “no undefined behavior” across all interactions—a formally undecidable property for open environments [15]. Replaced with closed interaction matrix coverage requirement.

Attack 3 — Threshold Arbitrariness (partially resolved): Thresholds marked SPECULATIVE (WSI-03, TC-03, SCE-01, SCE-02, EG-03) lack empirical calibration. SCE-04 receives analytical grounding via MVW (Section IV). Remaining thresholds are documented as open calibration tasks.

Attack 4 — Floating Point Non-Determinism (addressed): DR-01 cross-platform determinism is non-trivial because IEEE 754 behavior varies across compiler optimizations and CPU architectures [16], [17]. Investigation (Q3, Section VI) reduces this to a single hazardous operation in the kernel—now replaced by a portable cube root—and demonstrates a fixed-point fallback that is bit-identical across platforms by construction.

Attack 5 — Missing Observer Domain (resolved): The original seven domains constituted a formally closed system with no external verification mechanism. Domain 8 (Observer Interface) resolves this.

Net integrity assessment: 4 of 5 attacks fully resolved (the floating-point strategy now addressed via Q3), 1 partially resolved (threshold calibration, with SCE now empirically measured). The framework is structurally sound and now backed by a reference implementation.

Minimum Viable World: 6-tuple decomposition

$$\text{MVW} = (\mathbf{N}, \mathbf{M}, \mathbf{P}, \mathbf{K}, \mathbf{T}, \Phi) = (100, 8, 4, 3, 4, 1)$$

$\mathbf{N}$	$\mathbf{M}$	$\mathbf{P}$	$\mathbf{K}$	$\mathbf{T}$	Φ
= 100	= 8	= 4	= 3	= 4	= 1
entities	attributes {x,y,z, vx,vy,vz, m,type}	physics rules (Newton x3 + elastic)	spatial dims (Euclidean 3-space)	interactions {move,create, destroy,query}	observer endpoint

Fig. 2. The Minimum Viable World as a 6-tuple. Each parameter is the minimum value satisfying the Tier C definition; M , P , K , and Φ are VERIFIED from first principles, T is PLAUSIBLE, and N remains SPECULATIVE pending the emergence-threshold question (Q1).

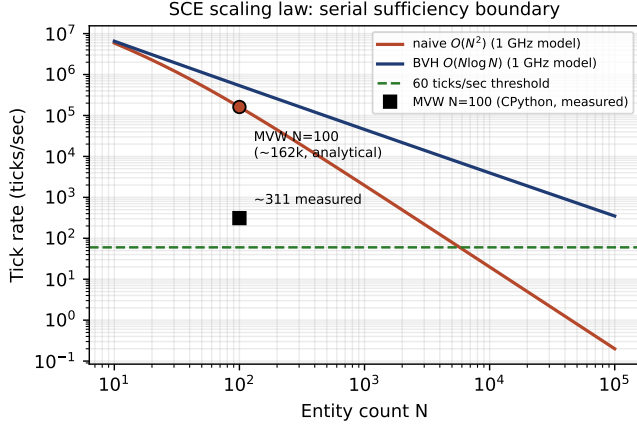


Fig. 3. SCE scaling law. The analytical 1 GHz model places naive $O(N^2)$ and BVH $O(N \log N)$ tick rates far above the 60 ticks/sec threshold at $N = 100$; the naive curve crosses the threshold near $N \approx 5,774$. The black marker is the empirical CPython reference measurement (~ 311 ticks/sec over a 60 s single-core run), which clears the threshold by $\sim 5\times$ despite a large interpreter tax. A profiling sweep confirms the empirical fits: naive = $O(N^2)$, BVH = $O(N \log N)$.

VI. OPEN RESEARCH QUESTIONS: INITIAL RESULTS

The four questions framed by this work were investigated with the reference implementation. Three are now resolved and one is partially resolved with empirical evidence; the experiments are reproducible from the repository.

Q1 — Minimum emergence threshold (partial): We operationalized “emergent behavior” with three proxies. (i) *Thermalization*: starting each particle at equal speed with random direction (each velocity component Uniform, excess kurtosis -1.2), an isolated elastic gas relaxes toward the microcanonical equilibrium, whose single-component excess kurtosis is exactly $-6/(3N + 2)$ —Gaussian (the Maxwell–Boltzmann

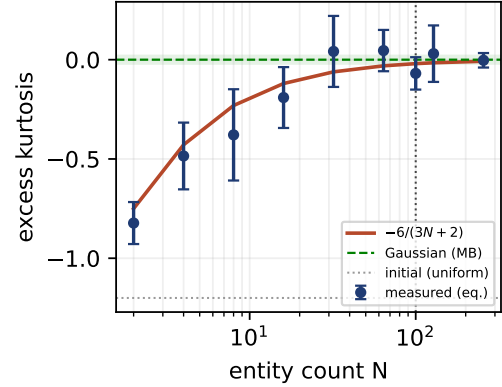


Fig. 4. Q1. Measured equilibrium velocity-component excess kurtosis vs. N (fixed density) tracks the microcanonical law $-6/(3N + 2)$, rising from the non-equilibrium initial value (-1.2) into the $\pm 2\%$ Gaussian (Maxwell–Boltzmann) band near $N = 100$.

signature) only as $N \rightarrow \infty$. Measured equilibrium kurtosis tracks this law (Fig. 4), giving a concrete threshold: the velocity distribution is Gaussian to within 2% at $N \approx 100$, within 5% at $N \gtrsim 40$. (ii) *Mixing*: a 10^{-6} perturbation diverges with positive Lyapunov exponent for all $N \geq 2$ —chaos is not the binding constraint. (iii) *Interaction-onset*: at the nominal box $L = 100$ the gas is collision-starved (~ 0 collisions/particle at $N = 100$). The binding under-specification is therefore *density*, not N : we recommend promoting the MVW to a 7-tuple with an explicit packing fraction. $N = 100$ is supported as a statistical minimum.

Q2 — BVH serial classification (resolved): The reference BVH build and traversal run in a single thread and are deterministic; the median-split construction has measured log–log time-scaling exponent ≈ 1.12 ($O(N \log N)$) versus ≈ 2.06 for

the naive scan. An independent serial broad-phase (sweep-and-prune) and the naive scan return identical overlap sets, so serial spatial indexing is not unique to BVH. Under Interpretation 1 the BVH *counts as serial*: it does not require parallelism, and the Section IV scaling analysis holds.

Q3 — Fixed-point vs. IEEE 754 (resolved): A static audit shows the kernel hot path uses only correctly-rounded IEEE-754 operations ($+$, $-$, $\times$, $\div$, $\sqrt{}$), which are bit-portable; the sole cross-platform hazard is the cube-root `pow` used once per entity for the collision radius, now replaced by a portable Newton-based cube root. Independently, a pure-integer fixed-point elastic-collision solver—bit-identical across platforms by construction—preserves momentum conservation to within 10^{-6} (the PSF-02 threshold) at scale factor $S \geq 2^{26}$, with error scaling as $1/S$. Cross-platform determinism is thus achievable either way.

Q4 — Turing completeness of interaction vocabulary (resolved): The subset $\{\text{create, destroy, query}\}$ is Turing-complete by reduction to a Minsky register machine [18]: a register is the population of an entity type, with $\text{INC} = \text{create}$, $\text{DEC} = \text{query} + \text{destroy}$, and $\text{JZ} = \text{query-for-zero}$. We confirmed this empirically by storing all machine state in an MVW world and executing register-machine programs (addition and multiplication, the latter with nested loops) through the interaction vocabulary alone; all returned correct results. The vocabulary is universal under query-conditioned control; `move` adds spatial expressiveness, not computational power, so $T = 4$ is sufficient (and not minimal) for the “arbitrary environment” claim of EG-01.

VII. CONCLUSION

We have presented a benchmark framework for holodeck-class simulation systems grounded in serial compute foundations, comprising eight domains, a formal Minimum Viable World definition, and a single-threaded standard-library reference implementation. An adversarial integrity analysis resolved four of five identified failures, the last (floating-point determinism) addressed through the research questions below.

The serial sufficiency claim is supported both analytically and empirically. The analytical model gives $\sim 162,000$ ticks/sec for a 100-entity Newtonian simulation at 1 GHz; the CPython reference engine sustains ~ 311 ticks/sec on a single core—still $\sim 5\times$ over the 60 ticks/sec threshold, the gap being interpreter overhead rather than an architectural limit. Advancing SCE-04 from measured to formally VERIFIED requires reproduction on the declared serial-baseline hardware.

The four research questions originally posed as open are now substantially answered: the emergent Maxwell–Boltzmann velocity distribution is Gaussian to within 2% near $N = 100$ (with spatial density, not N , identified as the binding parameter); the bounding-volume hierarchy is serial-compatible; both a portable cube root and a fixed-point fallback secure cross-platform determinism; and the interaction vocabulary is Turing-complete via a register-machine reduction verified in the reference engine. Three of the four are resolved; Q1 remains partially open on density calibration.

The benchmark set, reference implementation, and reproducible experiments are available as an open repository to support community validation efforts.

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